

1. Problem 1 (5.32)

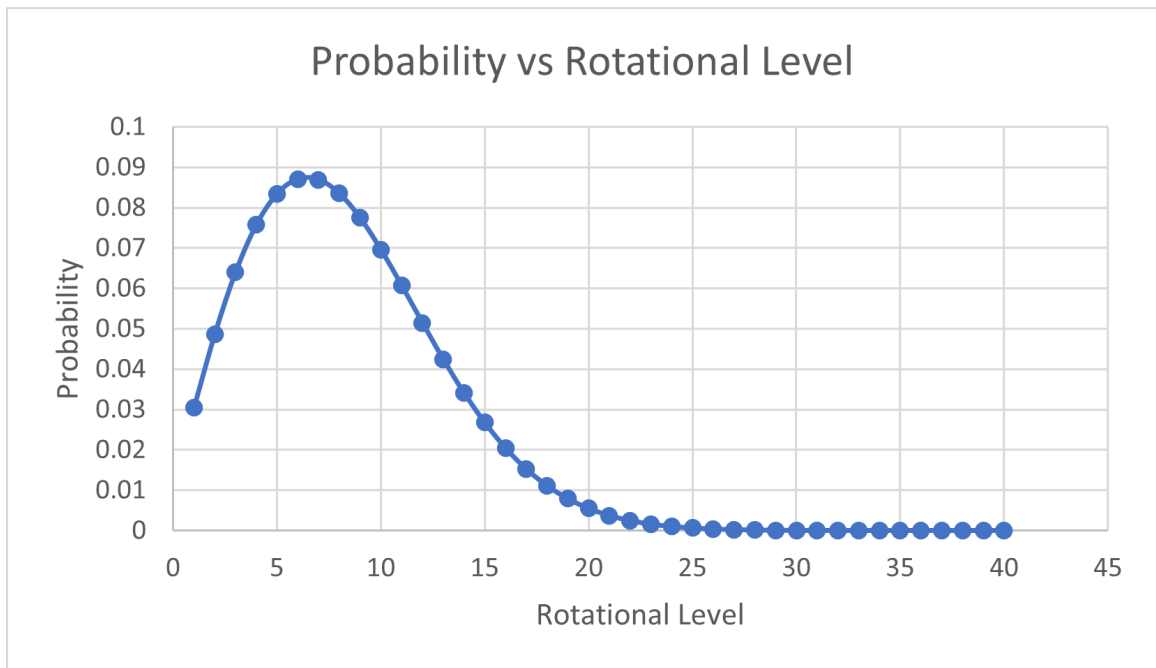
$$\sigma = 2$$

$$\theta_{\text{rot}} = 2.07$$

$$q_{\text{rot}} = \frac{T}{\sigma \theta_{\text{rot}}} = \frac{200}{2 \cdot 2.07} = 48.31$$

Plot:

$$P_J = \frac{(2J + 1) \exp\left(\frac{-J(J+1)\theta_{\text{rot}}}{T}\right)}{\sigma q_{\text{rot}}}$$



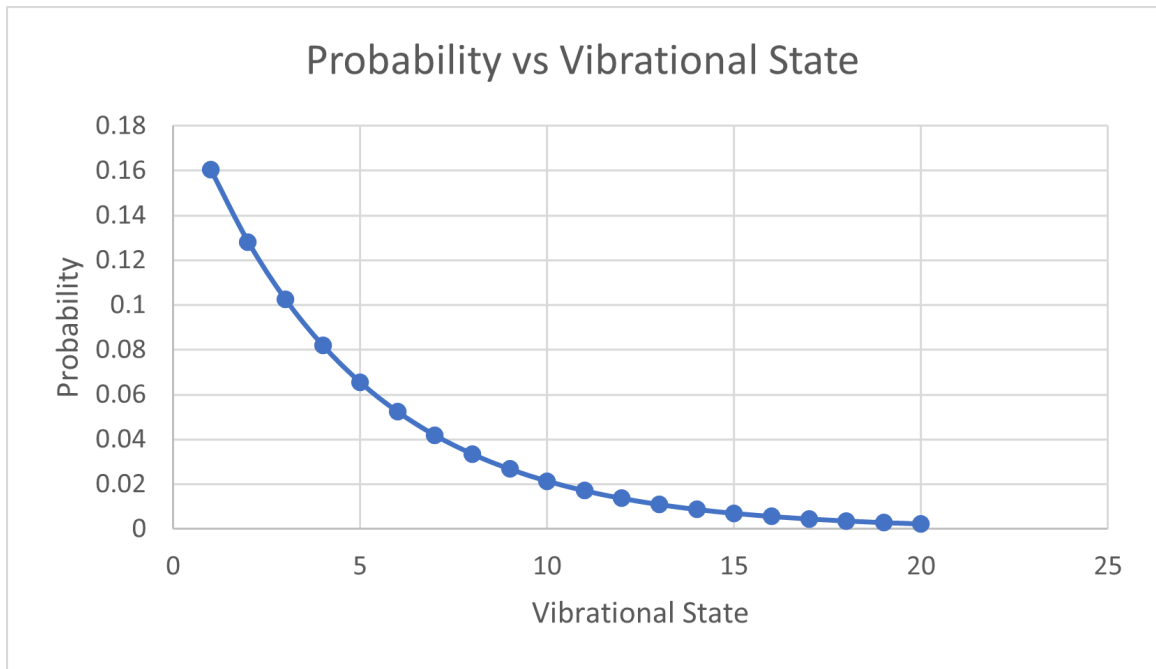
2. Problem 2 (5.33)

$$\theta_{\text{vib}} = 308$$

$$q_{\text{vib}} = \frac{1}{1 - \exp\left(\frac{-\theta_{\text{vib}}}{T}\right)} = \frac{1}{1 - \exp\left(\frac{-308}{1375}\right)} = 4.98$$

Plot:

$$P_n = \frac{\exp\left(\frac{-n\theta_{\text{vib}}}{T}\right)}{q_{\text{vib}}}$$



3. Problem 3 (5.34)

$$T = 1375$$

$$q_{\text{el, I}} = 4$$

$$q_{\text{el, I}_2} = 1$$

$$\theta_{\text{vib}} = 308$$

$$\theta_{\text{rot}} = 0.0537$$

$$D_0 = 148.8$$

$$\sigma = 2$$

$$\nu_{\text{I}_2} = -1$$

$$\nu_{\text{I}} = 2$$

$$P^\circ = 10^5$$

$$k_B = 1.3807 \cdot 10^{-23}$$

$$h = 6.626 \cdot 10^{-34}$$

$$m_{\text{I}} = 2.107 \cdot 10^{-25}$$

$$m_{\text{I}_2} = 4.215 \cdot 10^{-25}$$

$$\frac{k_B T}{P^\circ} = \frac{1.3807 \cdot 10^{-23} \cdot 1375}{10^5} = 1.898 \cdot 10^{-25}$$

$$q_{\text{rot}} = \frac{T}{\sigma \theta_{\text{rot}}} = \frac{1375}{2 \cdot 0.0537} = 12802.6$$

$$q_{\text{vib}} = \frac{1}{1 - \exp\left(\frac{-\theta_{\text{vib}}}{T}\right)} = \frac{1}{1 - \exp\left(\frac{-308}{1375}\right)} = 4.98$$

$$\exp\left(\frac{-\Delta U}{RT}\right) = \exp\left(\frac{-148.8 \cdot 100}{8.314 \cdot 1375}\right) = 2.22 \cdot 10^{-6}$$

$$K = \frac{k_B T}{P^\circ} \cdot \frac{\left(\frac{2\pi k_B T}{h^2}\right)^{3/2} \cdot m_{\text{I}}^3 \cdot q_{\text{el, I}}^2}{m_{\text{I}_2}^{3/2} \cdot q_{\text{rot}} \cdot q_{\text{vib}} \cdot q_{\text{el, I}_2}} \cdot \exp\left(\frac{-\Delta U}{RT}\right)$$

$$K = 1.898 \cdot 10^{-25} \cdot \frac{\left(\frac{2\pi \cdot 1.3807 \cdot 10^{-23} \cdot 1375}{(6.626 \cdot 10^{-34})^2}\right)^{3/2} \cdot (2.107 \cdot 10^{-25})^3 \cdot 4^2}{(4.215 \cdot 10^{-25})^{3/2} \cdot 12802.6 \cdot 4.98 \cdot 1} \cdot 2.22 \cdot 10^{-6}$$

$$K = 0.5128$$

a.

$$x = \left(\frac{K}{4\frac{P}{P^\circ} + K}\right)^{1/2}$$

$$x = \left(\frac{0.5128}{4\frac{0.1}{1} + 0.5128}\right)^{1/2}$$

$x = 0.7495$

b.

$$x = \left(\frac{0.5128}{4^{\frac{1}{1}} + 0.5128} \right)^{1/2}$$

$$\boxed{x = 0.3371}$$

c.

$$x = \left(\frac{0.5128}{4^{\frac{0.01}{1}} + 0.5128} \right)^{1/2}$$

$$\boxed{x = 0.9631}$$

4. Problem 4 (5.14)

a.

$$W_{A+B} = W_A W_B = 10^{26} \cdot 4 \cdot 10^{26}$$

$$\boxed{W_{A+B} = 4 \cdot 10^{52}}$$

b.

$$S_i = k_B \ln W_i$$

$$S_A = k_B \ln W_A = 1.3807 \cdot 10^{-23} \ln(10^{26})$$

$$\boxed{S_A = 8.26 \cdot 10^{-22} \text{ J/K}}$$

$$S_B = k_B \ln W_B = 1.3807 \cdot 10^{-23} \ln(4 \cdot 10^{26})$$

$$\boxed{S_B = 8.45 \cdot 10^{-22} \text{ J/K}}$$

$$S_{A+B} = S_A + S_B = 8.26 \cdot 10^{-22} + 8.45 \cdot 10^{-22}$$

$$\boxed{S_{A+B} = 1.67 \cdot 10^{-21} \text{ J/K}}$$

5. Problem 5 (5.40)

$$R \ln(3) = 1.0986R = 9.134 \text{ J/mol/K}$$

The difference between spectroscopic and calorimetric entropies is that NDH_2 has 3 rotational arrangements based on the position of the D atom.